End-to-End Test Plan — 22 Use Cases

Task 8.1 · v0.3 (plan) · Owner and test lead: Syed Sarim Jawad Zaidi · Based on SRS v1.0 §5 (NFR-01–13) and §6.2 (UC-01–22)

Goal: before the final demo, prove that every one of the 22 SRS use cases works end to end on real phones against one backend, and record how the system meets the 13 non-functional requirements. Sarim leads; Abdullah and Talha support and fix (roadmap Phase 8).

1. When the test starts and ends

	Rule
Start when	ICP-7 passed · all phases merged to dev · a release-candidate APK installed on the test phones · backend running (local or Render) with the scheduler on · test accounts and devices ready (section 3).
Finish when	22 of 22 use cases PASS (or an agreed N/A for an Optional feature) · no open Critical or High bug · every NFR has a recorded result · team signs the results document.
Pause when	A Critical bug blocks a whole use case (e.g. login broken, backend down): stop, log it, retest after the fix (8.4).

2. Who does what

Person	Role in 8.1
Sarim	Test lead: runs the plan, taps through the Dependent side, records results and bugs, keeps the daily bug list.
Abdullah	Backend: starts the server and short timers, reads server logs and the database, runs Postman checks, fixes backend bugs, runs NFR-01/02/06/13 measurements (8.2, 8.3).
Talha	App: builds and installs the APK, taps through the Guardian side, fixes app bugs, runs NFR-05 and NFR-12.

3. Test set-up

Item	Set-up
Phones	Phone A = Ammi (Dependent) · Phone B = Ali (Primary) · Phone C or emulator = Sara (Secondary) · Android 8.0 emulator for NFR-12
Accounts	Ammi and Nani (Dependents) · Ali (Primary of both) · Sara (alerts, communication, location, health records ON) · Bilal (all OFF) · Zain (verified Guardian, not linked) · emergency contact set
Backend	Latest dev; SCHEDULER_ENABLED=true; short timers for escalation days (1-minute window and levels, as in the 3.17 document); full Postman collection passes first
Hardware	Wearable band charged and paired (UC-17) · spare phone in camera mode in a room (UC-13)
Network	Normal Wi-Fi for most cases; 4G on Phone A and B for NFR-01

4. The 22 use cases

Each row is one end-to-end run across both apps. The detailed case documents already in docs/qa give the exact steps; this table is the order and the pass rule. "Backend now" shows what exists on dev today (22 Oct 2026, up to PR #80): all 22 use cases have their backend (UC-17 with mock band data; UC-13 video needs the camera phone). App screens exist for every use case except calls (4.7) and the camera phone mode (6.10); see the Full Task List.

UC	Use case	Actors	FR	End-to-end run	Pass when	Detailed cases	Backend now	Result
UC-01	Register as Guardian or Dependent	Guardian, Dependent	FR-A1–A3	Ali registers and verifies (OTP + CNIC). Ammi registers, passes KYC (CNIC + live photo), cannot log in until linked.	Accounts created; roles fixed; Ammi blocked until verified AND linked.	1.13 / 2.9 F-01–F-06	Built	
UC-02	Link Dependent to Primary Guardian	Primary, Dependent	FR-B1, B2, B4	Ammi shows her ID (QR); Ali links her; a second Guardian tries to become Primary; Ali also links Nani.	Link works; second Primary refused; Ali sees both Dependents and switches between them.	2.9 F-07, F-08, F-10	Built	
UC-03	Invite Secondary Guardian	Primary	FR-B1, B2	Ali invites Sara and Bilal; tries a 4th Guardian.	Both join as Secondary; 4th refused (1 + 2 cap).	2.9 F-09	Built	
UC-04	Configure permission dashboard	Primary	FR-A4	Ali grants Sara alerts + communication + location, Bilal nothing; Sara tries to grant herself.	Each Secondary sees only granted modules; self-grant refused; change shows at once.	2.9 F-11–F-13	Built	
UC-05	View live location	Guardian	FR-C1, C2, C4	Ammi walks around with her phone; Ali and Sara open the Live Map and history.	Position updates (10 s moving / 60 s still); Bilal (no permission) refused; route shown for the day.	2.8 mockups 11, 11c	Built	
UC-06	Configure geofence(s)	Primary	FR-C2, C3	Ali adds Home and Mosque zones; Ammi leaves both.	One amber breach alert per exit; closes on return; max 10 zones.	2.8 mockups 12, 12b	Built	
UC-07	Configure medication schedule	Guardian	FR-F1	Ali adds Metformin 3 times a day.	Schedule saved; Ammi gets a reminder 15 min before each dose.	3.17 E-01	Built	
UC-08	Confirm medication taken	Dependent	FR-F1, D1	Ammi confirms one dose in two steps; skips the next.	Taken recorded; skipped dose becomes missed and alerts Ali.	3.17 E-01, E-06–E-09	Built	
UC-09	Configure Wellness Check schedule	Guardian	FR-D4	Ali applies the default schedule, then edits one time and grace period.	3 checks a day, grace 5–180 min, saved in Karachi time.	3.17 E-10	Built	
UC-10	Respond to Wellness Check	Dependent	FR-D4	Ammi answers one check; ignores one with no movement; ignores one while walking.	Answered closes; silent + still escalates; silent + moving closes quietly.	3.17 E-10–E-12	Built	
UC-11	Trigger Panic Mode	Dependent	FR-D2	Ammi holds Panic for 2 s; presses twice; then says she is safe.	Every Guardian alerted at once (even Bilal); double press = one panic; all-clear sent.	3.17 E-14–E-17	Built	
UC-12	Trigger Remote Alarm Override	Primary	FR-D3	Ali rings Ammi's phone on silent; Sara tries.	Phone rings loudly for the set time; Sara refused; Stop works.	3.9 mockups 14, 24	Built	

UC	Use case	Actors	FR	End-to-end run	Pass when	Detailed cases	Backend now	Result
UC-13	View live camera feed	Guardian	FR-K1, K2	Ali views the room camera; Ammi presses Panic.	Live video; Ammi's phone shows the viewing notice every time; clip saved on Panic.	6.12 K-01–K-17	Built (6.3, 6.4); video needs the camera phone (6.10)	
UC-14	Communicate (message / call)	Guardian, Dependent	FR-E1, E2, E4	Messages both ways, an urgent voice message, a voice and a video call.	Delivered once, read receipts, critical alert for voice, call answered and ended.	4.10 M-, V-, C-	Built	
UC-15	Manage medical records	Guardian, Dependent	FR-G1–G4	Upload records, restrict Sara to lab reports, export a month PDF, add an appointment.	Right people see the right records; PDF correct; both reminded.	5.10 R-01–R-16, RA-01–RA-09, AP-01–AP-05	Built (5.1, 5.2)	
UC-16	Record daily vitals	Guardian, Dependent	FR-H1	Ammi enters BP; Ali enters sugar, SpO2, temperature.	Saved with who and when; numbers only, never high / low; impossible values refused; vitals PDF correct.	5.10 H-01–H-12	Built (5.3)	
UC-17	Sync wearable data	Dependent, Backend	FR-H2	Pair the band; sync; walk out of range and back.	Heart rate appears; after 30 min without data the "band not sending" notice; reconnects when back; nothing else blocked.	5.10 W-01–W-10	Mock data (5.4); real band 5.5	
UC-18	Browse healthcare directory	Guardian, Dependent	FR-I1, I2	Search, filter, call an ambulance service.	Correct number dialled; call logged in the activity log.	6.12 D-01–D-08, D-15, D-16	Built (6.1, 6.2)	
UC-19	Request home-care service	Guardian, Dependent	FR-J1, J2	Request a nurse visit; status changes.	Pending → Confirmed → Completed in the activity log; no payment step.	6.12 D-09–D-18	Built (6.2)	
UC-20	View analytics dashboard	Guardian	FR-Q1, Q2	Open adherence and vitals analytics after a week of test data.	Figures match the raw data (doses taken / missed, vitals trends); reports PDF matches; numbers only.	7.7 F-15 (API); app cases to write	Built (7.1, 7.2)	
UC-21	Receive emergency escalation	Secondary, External contact	FR-D5, R1	Let a missed dose and a missed check go unopened.	Primary → Secondary (5 min) → one SMS (5 min); stops when opened.	3.17 E-02–E-05, E-13, E-18–E-21	Built	
UC-22	View audit log	Primary	FR-N3	Change permissions, view camera and records, then open the log.	Every sensitive action listed with who, what and when.	6.12 K-10, K-15, 5.10 R-10	Built (6.6)	

5. Non-functional checks

NFR	What	How we test it	Pass when	Who	Result
NFR-01	Command latency	Stopwatch from Panic / Remote alarm press to the other phone reacting, on 4G, 10 tries.	All 10 within 2 s	8.2 Abdullah, recorded by Sarim	
NFR-02	Battery	Dependent phone idle with location on for 2 hours, screen off; read battery use per hour.	5% per hour or less	8.2 Abdullah	
NFR-03	Call encryption	Confirm with Abdullah how ZegoCloud encrypts audio / video; note the setting used.	Encrypted end to end, documented	Abdullah	
NFR-04	Data at rest	Confirm how records, vitals and location are encrypted in the database / storage.	Method documented and switched on	Abdullah	
NFR-05	Availability / queuing	Stop the backend; press Panic on Ammi's phone; start the backend again.	Panic is sent as soon as the server is back	Talha	
NFR-06	3 Guardians	Ali, Sara and Bilal all watch Ammi's map and alerts at the same time.	No slowdown or missed alerts	8.2 Abdullah	
NFR-07	2-tap rule	Count taps from Dependent home to Panic, I am OK, confirm medicine.	Each within 2 taps	Sarim (3.16)	
NFR-08	Contrast	Measure text contrast on the 3 safety screens with Large Text / High Contrast on.	4.5 : 1 or better	Sarim (3.16)	
NFR-09	Urdu	Switch to Urdu and visit every screen.	No English text left; right-to-left correct	Sarim	
NFR-10	Privacy	Check each screen and API reply shows only needed data; permissions respected.	No extra personal data	Sarim	
NFR-11	Maintainability	Each module's tests run on their own (backend test suite by module).	Pass per module	Abdullah	
NFR-12	Android range	Install and smoke-test on Android 8.0 (API 26) emulator and the newest phone available.	Works on both	Talha	
NFR-13	Band reconnect	Walk out of range with the band, come back, time the reconnect, 5 tries.	Within 30 s every time	8.3 Abdullah	

6. Proposed schedule (Phase 8)

Day	Covers
Day 1	UC-01 to UC-06 (accounts, linking, permissions, location, safe zones) + NFR-12 install check
Day 2	UC-07 to UC-12 and UC-21 with short timers (medicine, wellness, Panic, alarm, escalation) + NFR-01, NFR-05, NFR-07
Day 3	UC-13 to UC-20 and UC-22 (communication, records, vitals, band, directory, camera, analytics, audit log)
Day 4	Remaining NFRs (02, 06, 08, 09, 13), retest of fixed bugs, results document and sign-off

7. Bugs and results

- Every failure gets a bug ID (BUG-001, …) with steps, expected, actual, a screenshot, severity and owner, using the 1.13 checklist's bug log and severity table.
- Critical and High bugs are fixed and retested before sign-off (8.4 bug-fixing sprint); Medium and Low may wait for polish.
- Results go in a copy of the QA checklist: Humsafar_QA_8.1_E2E_Results_v1.0 (Word in Drive 02_QA, PDF in docs/qa).

8. Open points

- UC-20 (analytics): the API is covered by the 7.7 regression run (F-15); app-side cases to be written with Talha's 7.3 screen once its field names are fixed (6.13 F-01).
- Analytics FR-Q2 summary reports (Optional) are built (7.2) and tested in UC-20. The FR-G3 doctor link (Optional) is built (5.1), so it is tested in UC-15.
- UC-17: step count is not built (waits for the band's sensor list, drift D-33); test heart rate only until then. Real-band runs wait for Task 5.5.
- UC-18 and UC-19: the directory holds demo entries (marked Sample); replace them with real, checked entries before the final run (drift D-35).
- v0.2 changes: Backend now column and linked cases updated for Tasks 5.1–5.4 and 6.1 (UC-15 to UC-19).
- v0.3 changes: all 22 use cases now have their backend (6.2–7.2); UC-13, UC-18, UC-19, UC-20, UC-22 updated; FR-K1.3 notice in the app needs Talha's field-name fix before UC-13 can pass.
- NFR-03 and NFR-04 are confirmed by Abdullah's documentation rather than by a tap-through.